

Roman Penunuri

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SUMMARY

Fourth-year Electrical Engineering student concentrating in Power Systems, with industry experience in industrial electrical asset management and condition-based maintenance, as well as direct project experience analyzing solar energy storage economics across global regions. Familiar with the operational and economic tradeoffs between storage technologies and the factors driving grid integration decisions at scale. Pursuing formal coursework in Power System Analysis and active self-study in power electronics through MIT OpenCourseWare and Coursera. Self-motivated and driven to see technical problems through from start to finish, proactive about seeking input, and comfortable integrating ideas across disciplines.

EDUCATION

University of Houston, Cullen College of Engineering

Houston, TX

B.S. Electrical Engineering, Power Systems Concentration | GPA: 3.82

May 2028

Dean's List: Fall 2024, Spring 2026

Relevant Coursework: Power System Analysis (in progress), Circuits and Systems, Numerical Methods, Engineering Statistics, Signals & Systems Analysis, Applied Electromagnetic Waves (Honors), Microprocessor Systems

SKILLS

Certifications: OSHA-30 General Industry | SEL CBT 101 Protective Relay Fundamentals (4 PDH) | Pursuing ETAP Operator Certification

Proficient: MATLAB, GE Vernova EnerVista (Multilin), Maximo CMMS, MS Suite, Google Workspace

Currently Learning: ETAP, AutoCAD Electrical, Power Electronics (MIT OCW and Coursera), Python

AI Tools: Proficient in AI-assisted development workflows for rapid prototyping and data tool development

Languages: Fluent in English, Conversational in Spanish

PROJECTS

NAE Grand Challenge: Solar Energy Storage Dashboard

University of Houston

- Led integration and built the solar economics module for a MATLAB App Designer interactive tool comparing energy storage technologies and solar infrastructure economics across global regions; analyzed storage performance metrics, cost factors, and grid integration tradeoffs to support infrastructure decision-making

WORK EXPERIENCE

Air Liquide

Pasadena, TX

Plant Operations Electrical Engineering Intern

June 2025 - August 2025

- Implemented condition-based maintenance protocols for industrial electrical assets (motors, switchgear, batteries, UPS systems) by revising NFPA 70B- and IEEE-compliant documentation; reduced unnecessary maintenance cycles and extended equipment lifecycle across multiple facilities
- Upgraded motor protection relay (GE Vernova Multilin 469 to 869) by migrating and validating all settings in EnerVista, eliminating a recurring fault source and reducing projected maintenance costs by ~\$10,000
- Authored three scope-of-work packages for a plant turnaround covering electrical equipment, battery systems, and motor assets; coordinated with Engineering, Safety, and Maintenance to mitigate potential operational losses exceeding \$1M
- Developed CT wiring schematics and scope-of-work for industrial bus heater ammeter monitoring, enabling real-time current measurement to detect failures before bus duct damage
- Used Maximo CMMS to audit equipment maintenance histories, identify PM-due assets, and reconcile turnaround planning workbooks
- Selected for return internship (Summer 2026) based on performance and documented project impact

Take 5 Oil Change

Crosby, TX

Certified Lube Technician

March 2025 - May 2025

- Executed 40+ vehicle service cycles daily to manufacturer specifications in a fast-paced, safety-critical environment, maintaining procedural discipline across high-volume operations
- Diagnosed service needs and translated technical findings into clear customer recommendations through structured verbal communication

Planet Fitness

Crosby, TX

Member Services Representative

January 2024 - March 2025

- Managed operations for a 1,000+ member facility, maintaining equipment, cleanliness standards, and a >90% satisfaction rating, while converting an average of 6 prospects per week through structured membership presentations

ORGANIZATIONS

Tau Beta Pi Honor Society

Houston, TX

Member

April 2025 - Present

Society of Hispanic Professional Engineers (SHPE)

Houston, TX

Member

August 2023 - Present

IEEE University of Houston Chapter

Houston, TX

Member

August 2025 - Present